Luben M. C. Cabezas

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Luben Miguel Cruz Cabezas | Monoxido45

Education

Phd — Aug 2022 – Present

Federal University of São Carlos, Statistics
ICMC - University of São Paulo, Statistics

- **GPA:** 4.0/4.0
- **Relevant Coursework:** Advanced Probability, Advanced Inference, Statistical Learning, Advanced Topics in Statistical Learning, Stochastic Simulation
- **Fellowship:** Funded by FAPESP (Grant 2022/08579-7)

Exc. Phd — Sep 2025 – Sep 2026

Inria Grenoble, Statistics and Machine Learning

- **Part of the project:** Simulation Based Inference for Climate (SBI4C)
- **Fellowship:** Funded by FAPESP (Grant 2025/06168-8)

BS — Mar 2018 – Apr 2022

Federal University of São Carlos, Statistics

- GPA: 9.24/10
- **Relevant Coursework:** Probability, Statistical Inference, Bayesian Inference, Machine Learning, Regression analysis, Stochastic Processes, Computationally Intensive Methods

Working Experience

Statistical Machine Learning Lab - SMaLL (UFSCar), Researcher — São Carlos, SP, Brazil, Aug 2022 – Present

- Ph.D. researcher at SMaLL, supervised by Rafael Izbicki and Rafael B. Stern.
- Research focused on conformal prediction, likelihood-free inference, nonparametric inference, and hypothesis testing.
- Developed *LOCART*, a tree-based locally adaptive conformal method, published in Information Sciences.
- Developed *EPICSCORE*, a method that uses Bayesian modeling to add epistemic uncertainty to any conformal score, published at UAI 2025 (oral).
- Designed and maintained open-source Python and R packages implementing these methods.
- Research funded by FAPESP (Grant 2022/08579-7).

Statify (Inria), Visiting Researcher — Grenoble, France, Sep 2025 – Sep 2026

- Visiting Ph.D. researcher at Statify, supervised by Pedro L. C. Rodrigues, within the SBI4C project.
- Research focused on simulation-based inference, climate and ocean model calibration, ensemble Kalman methods, and conformal prediction.
- Developed *CP4SBI*, a conformal method for locally calibrating credible regions in SBI, published in Philosophical Transactions of the Royal Society A.
- Workshop papers at NeurIPS: SBI for ocean turbulence model calibration and posterior estimation (Sim2Science workshop, NeurIPS, 2026) and three-way testing for epistemic uncertainty in SBI (EIML workshop, EurIPS, 2025).
- Research funded by FAPESP (Grant 2025/06168-8).

Terranova (Brazilian Jurimetrics Association), Statistics Intern — São Paulo, SP, Brazil, Nov 2021 – June 2022

- Developed and maintained internal R-based packages for data analysis and modeling.
- Worked on proof-of-concept (POC) development for predictive models.
- Contributed to the implementation and debugging of Shiny dashboards.

Statistical Machine Learning Lab - SMaLL (UFSCar), Research Assistant — São Carlos, SP, Brazil, Nov 2020 – Nov 2021

- Undergraduate researcher at SMaLL, supervised by Rafael Izbicki.
- Developed visualization, feature-importance, and scoring methods for hierarchical clustering via phylogenetic probabilistic models, published in Applied Soft Computing (2023).
- Research funded by FAPESP (Grant 2020/10861-7).

Publications and pre-prints

LoBoost: Model-Native Local Conformal Prediction for Gradient-Boosted Trees — Nov 2026
Vagner S. Santos, Victor A. Coscrato, **Luben M. C. Cabezas**, Rafael Izbicki, Thiago R. Ramos
Published in Transactions on Machine Learning Research (TMLR)

Conformal Prediction via Transported Beta Laws — Sep 2026
Thiago R. Ramos, Helton Graziadei, **Luben M. C. Cabezas**
Published in 15th Symposium on Conformal and Probabilistic Prediction With Applications (COPA) PMLR 329:133-168, 2026.

CP4SBI: local conformal calibration of credible sets in simulation-based inference — Aug 2026
Luben M. C. Cabezas, Vagner S. Santos, Thiago R. Ramos, Pedro L. C. Rodrigues, Rafael Izbicki
Published in Philosophical Transactions of the Royal Society A 10.1098/rsta.2025.0069

Conformal Calibration of Statistical Confidence Sets — Apr 2026
Luben M. C. Cabezas, Guilherme P. Soares, Thiago R. Ramos, Rafael B. Stern, Rafael Izbicki
Published in Transactions on Machine Learning Research (TMLR)

Epistemic Uncertainty in Conformal Scores: A Unified Approach — Jun 2025
Luben M. C. Cabezas, Vagner S. Santos, Thiago R. Ramos, Rafael Izbicki
Published in the 41st Conference on Uncertainty in Artificial Intelligence (UAI) PMLR 286:443-470, 2025

REACT to NHST: Sensible conclusions to meaningful hypotheses — May 2025
Rafael Izbicki, **Luben M. C. Cabezas**, Fernando A. B. Colugnatti, Rodrigo F. L. Lassance, Altay A. L. de Souza, Rafael B. Stern
Published in The Quantitative Methods for Psychology (TQMP) 10.20982/tqmp.21.2.p043

Regression trees for fast and adaptive prediction intervals — Jan 2025
Luben M. C. Cabezas, Mateus P. Otto, Rafael Izbicki, Rafael B. Stern
Published in Information Sciences (IS) 10.1016/j.ins.2024.121369

Hierarchical Clustering: visualization, feature importance and model selection — July 2023
Luben M. C. Cabezas, Rafael Izbicki, Rafael B. Stern
Published in Applied Soft Computing (ASoC) 10.1016/j.asoc.2023.110303

CREDO: Epistemic-Aware Conformalized Credal Envelopes for Regression (pre-print) — Mar 2026
Luben M. C. Cabezas, Sabina J. Sloman, Bruno M. Resende, Fanyi Wu, Michele Caprio, Rafael Izbicki
arxiv:2603.06826

Conformal Prediction for Compositional Data (pre-print) — Nov 2025

Lucas P. Amaral, **Luben M. C. Cabezas**, Thiago R. Ramos, Gustavo H. G. A. Pereira
arxiv:2511.18141

Teaching Experience

Teaching Assistant, Department of Statistics, UFSCar. Assisted in the course "Statistical Inference" for second-year Statistics students (90 hours total) during the second semester of 2022 (class 1001736). Responsible professor: Rafael Izbicki.

Teaching Assistant, Department of Statistics, UFSCar. Assisted in the course "Data Mining" for third-year Statistics students (60 hours total) during the first semester of 2022 (class 158518). Responsible professor: Rafael Izbicki.

Supervisions

Aurèle Gallard (M2 internship, co-supervised with Pedro L. C. Rodrigues and Florence Forbes) — May 2026 – Oct 2026
École Centrale Paris – Inria Grenoble Rhône-Alpes
Project: Calibration of subgrid parametrizations in large-scale climate models

Awards

Best Paper Award, 15th Symposium on Conformal and Probabilistic Prediction with Applications (COPA 2026) — Sep 2026
Conformal Predictio via Transported Beta Laws, with T. R. Ramos and H. Graziadei

Oral Presentation, 41st Conference on Uncertainty in Artificial Intelligence (UAI 2025) — Jul 2025
Epistemic Uncertainty in Conformal Scores: A Unified Approach, selected for oral presentation

Outreach and Leadership

Reviewer for EurIPS 2025 workshop: Epistemic Intelligence in Machine Learning.
Reviewer for UAI 2026 main conference.
Reviewer for NeurIPS 2026 workshop: Sim2Science: ML with Imperfect Scientific Models.

Work Presentations

15th Symposium on Conformal and Probabilistic Prediction With Applications (COPA): Conformal Prediction via Transported Beta Laws, 2026. Gothenburg, Sweden.

Second Edition of the Grenoble AI for Physical Sciences, AISSAI-GAP 2026: CP4SBI: Local Conformal Calibration of Credible Sets in Simulation-Based Inference, 2026. Grenoble, France.

Statlearn 2026: Epistemic Uncertainty in Conformal Scores: A Unified Approach, 2026. Grenoble, France.

1st Epistemic Intelligence in Machine Learning at EurIPS 2025: A Three-Way Testing Framework for Quantifying Epistemic Calibration Uncertainty in SBI, 2025. Copenhagen, Denmark.

21st School on Time Series and Econometry (ESTE): Distribution-Free Calibration of Statistical Confidence Sets, 2025. Campinas, SP, Brazil.

41st Conference on Uncertainty in Artificial Intelligence: Epistemic Uncertainty in Conformal Scores: A Unified Approach, 2025. Rio de Janeiro, RJ, Brazil.

VII Latin American Meeting on Bayesian Statistics (COBAL) and XVII Brazilian Meeting of Bayesian Statistics (EBEB): Pragmatic variable selection in nonparametric binary regression, 2024. Belo Horizonte, MG, Brazil.

X Workshop on Probabilistic and Statistical Methods (WPSM): Regression Trees for Fast and Adaptive Prediction Intervals, 2024. São Carlos, SP, Brazil.

67ª RBras e 20º SEAGRO: Toward Local and Valid Uncertainty Estimation in ML, 2023. Londrina, PR, Brazil.

XVI CLAPEM (Latin America Congress of Probability and Mathematical Statistics) Towards Local and Valid Uncertainty Estimation in ML, 2023. São Paulo, SP, Brazil.

XXVIII Congresso de Iniciação Científica e XIII Congresso de Iniciação em Desenvolvimento Tecnológico e Inovação (UFSCar) A data-splitting approach for comparing hierarchical clustering algorithms, 2022. São Carlos, SP, Brazil (online).

XXVII Congresso de Iniciação Científica e XII Congresso de Iniciação em Desenvolvimento Tecnológico e Inovação (UFSCar): Previsão em modelos filogenéticos, 2020. São Carlos, SP, Brazil (online).

Skills

Coding Languages

- Python, R, Shell Script, SQL

Developing tools:

- Git, Markdown, Linux, Latex

Languages

- Portuguese: Native
- English: Fluent
- Spanish: Advanced
- French: Upper-Intermediate